Water Research 221 (2022) 118850

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

Upgrade the high-load anaerobic digestion and relieve acid stress through the strategy of side-stream micro-aeration: biochemical performances, microbial response and intrinsic mechanisms

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ARTICLE INFO

Keywords:
High-load anaerobic digestion
Acid stress
Side-stream micro-aeration
Metagenomic analysis

ABSTRACT

In high-load anaerobic digestion such as in kitchen waste, side-stream micro-aeration (SMA) shows excellent operational performance to direct micro-aeration (DMA). It immediately restores the acidification to stability. Methanogenic performance remained stable when organic load ratios (OLR) was further increased to 5.5 g VS/L. Enhanced enzyme activity, microbial aggregation, and proliferation of bacteria and archaea were observed in SMA. The results indicates that SMA enriched *Methanosaeta* (relative abundance exceeded 93%) and induced the change of the main methanogenic pathway to acetoclastic methanogenesis. Mechanisms was further explored by using metagenomic analysis, and the results show SMA avoids mass formation of ROS (reactive oxygen species) by cycling the aerated slurry, and retains benefits of trace O_2 on material and energic metabolism, which poses great application potentials and deserves further investigation.

1. Introduction

Urban food waste (FW) is an important part of household waste that is rich in nutrients, which can be utilized via anaerobic digestion (AD). However, the high proportion of easily biodegradable organic matter in FW commonly results in rapid acidification, as well as subsequent acidity accumulation in its AD, especially when high feed load is applied. The acidity accumulation deteriorates systematic stability by inhibiting anaerobic microbes, and if not properly treated, system failure occurs. However, AD with high organic load ratios (OLR) presents the advantage of high volume gas yield and small reactor volume, which have great potential for AD-engineered treatment of different kinds of organic wastes (Li et al., 2022). Therefore, it is of great significance to effectively maintain the stability of such a AD system. Effectively diluting the adverse effects of inhibitors during high-load anaerobic digestion is a key hurdle.

Several studies have explored to handle acid inhibition, including the use of exogenous alkaline (Chen et al., 2019), cultivating acid-resistant strains via pre-inoculation, gradient culture, or using trace elements to enhance microbial activity (Zhang and Loh, 2019), and accelerating the transfer of electrons from syntrophic oxidation bacteria to methanogens through exogenous conductors (Zhao, 2020). In these processes, the fundamental goal is to effectively reduce the accumulation of fatty acids, such as propionate and butyric acid, which cannot be directly used by methanogenic archaea.

In recent years, micro-aeration has attracted much attention in ameliorating these challenges. In such processes, a small amount of air is injected into the anaerobic digestive system to induce the fluctuation of the digestive ORP (Oxidation Reduction Potential). Trace oxygen acts as an electron receptor to capture/shunt excess electrons produced in AD of perishable organics, alleviating acid inhibition and promoting methanogenesis. Yu et al.(2020) observed that during AD of black water, an appropriate micro-oxygen supply improved hydrolysis, acidification and even methanogenic efficiency, and the abundance of hydrogenotrophic methanogens significantly increased. According to Nguyen et al.(2019), intermittent micro-aeration rapidly decreased the total VFA concentration in the reactor from 9.2 to 3.0 g HAc/L and induced the recovery of methane production, thereby improving overall stability of the AD system.

Recently, direct introduction of air into the digester is the most common form of intervention used. This kind of micro-aeration is impractical for engineering application due to the high cost of full-scale

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https://doi.org/10.1016/j.watres.2022.118850
Received 16 April 2022; Received in revised form 3 July 2022; Accepted 9 July 2022
Available online 11 July 2022
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digesters emptying and reforming, as well as difficulties in assuring the homogeneous diffusion of air bubbles to avoid too high ORP in certain areas. Compared with direct micro-aeration, side-stream micro-aeration (SMA) has great potential for application because micro-aeration can be operated in a side-stream reactor with a much smaller volume, which avoids heterogeneous diffusion of air in the main digester and lowers the potential risks in systems reforming and subsequent operation. Gonzalez-Salgado et al. (2020) connected a thermophilic aerobic reactor to mesophilic anaerobic digestion by means of a peristaltic recirculation pump to circulate the slurry. This side-stream form somewhat increased the efficiency of hydrolysis and organics removal. However, research about side-stream micro-aeration remains limited, especially in regard to performance and mechanism in facing acidity stress.

Here, the side-stream micro-aeration approach for AD in FW under mesophilic conditions is adopt, aiming to relieve acidity stress in order to maintain systematic stability under high OLRs. Digestion performance under different reactor modes were examined and compared. This work focused on the ability of the side-stream micro-aeration to relieve acid inhibition. To clarify the mechanism of this kind of micro-aeration in relieving acidity stress, we investigated the response and feedback of the microbial community structure, as well as the microbial metabolic pathway. The results of this study provide new insights and reference for improving the stability of overloaded AD systems.

2. Materials and methods

2.1. Substrates and inoculum

Inoculum applied in this study was taken from an anaerobic digestion tank of Tangjiatuo wastewater treatment plant (WWTP) in Chongqing, China. The WWTP adopts A^2/O process and the treatment capacity is about 40,000 t/d. The inoculation sludge was pretreated by settling and subsequent filtration through a 20-mesh screen to get purified and concentrated. The prepared inoculum was stored under anaerobic conditions at 35±1 °C for more than one week before use to reduce background methanation. The substrate used in this work is simulated FW to avoid the unnecessary effect of organic fractions' variation on digestive efficiency. The simulated FW is composed by 400 g rice, 300 g cabbage, 200 g pork and 100 g tofu. These raw materials were firstly cooked in distilled water, and then were crushed to assure the particle size less than 5 mm. Finally, the crushed slurry was diluted to the volume of 2.0 L. Characteristics of the substrates and the inoculum are listed in Table S1.

2.2. Digester set-up and operational conditions

The side-stream micro-aeration system (SMA) consisted of a main digester and a side-stream reactor equipped with aeration devices (aerator and gas tube). Their volume were 3.0 and 0.7 L. Schematic diagram of the device structure can be found in Fig. S1. The AD system was equipped with feed hoppers, a mixing shaft, a separate jacketed thermostat, ORP online monitoring, and gas collection devices. Digests were excreted from the main reactors. Daily circulation amount of the slurry from the main digester to the side-stream reactor accounted for 18% of the total by referring from previous study (Wu et al., 2016). The recirculation was realized by a peristaltic pump with a flow rate of 190 mL/min. The operation was as follows: 0.7 L of sludge from the side-stream reactor was completely transferred to the main reactor and mixed thoroughly with the slurry in the main reactor. Another same system without aeration devices (SD), as well as a direct micro-aeration digester without side-stream reactor (DMA) were also built to provide comparison. The experimental design and the analysis performed was described in Table S2a. All three AD systems were operated under a semi-continuous feeding mode to maintain a constant hydraulic retention time (HRT) of 20 days and the OLRs ranging from 2.5 to 5.5 g VS/L·day (shown in Table S2b). The digestive temperature was maintained at 35±1 °C. According to the OLRs, the experimental process can be divided into five stages. In fourth and fifth stage, a little air was introduced daily after substrate feeding. The dose was 20 mL and the aeration intensity was 0.03L Air/L reactor.d. And this was selected as the optimal dose by pre-experiments (Fig. S2). Air was injected into the side-stream digester in a feeding-discharge period through a syringe over 10 min and the injection process was manually controlled to proceed uniformly and slowly, resulting in the injection rate of approximately 2 mL/min. All reactors and experiments were triply operated.

2.3. Analytical methods

2.3.1. Chemical and physical analysis

Measurements of total solid (TS), volatile solid (VS), chemical oxygen demand (COD), alkalinity and NH_4^+-N were conducted in accordance with standard methods (Gao et al., 2022). pH was determined by pH meter (FE28, METTLER TOLEDO, Switzerland). Gasbags were used to collect the biogas in the digesters, and the volume was measured by a graduated syringe through repeat extracting the gas from the bag. The biogas composition and volatile fatty acids (VFAs) were determined by a gas chromatography (Jiedao GC1690, Hangzhou). The operation process of gas chromatography was introduced in detail in previous study (Tian et al., 2021). α-Glucosidase activity, ATP content, coenzyme F_{420}, CAT (catalase), as well as acetate kinase were measured by using the ELISA kit (Shanghai hengyuan biological technology co., LTD). Electron transfer activity (ETS) was measured using the method reported by Wan et al. (2016). Quantification of C, H and O elements using an elemental analyzer (Elementar Analysensysteme GmbH, Unicube). Confocal laser scanning microscope (CLSM) staining and observation was conducted according to the method reported by Chen et al. (2007).

2.3.2. Microbial community and absolute quantitative PCR analyses

Samples were taken from the raw sludge and SMA digester at the end of stage III and IV for microbial analysis. Microbial communities were analyzed by high-throughput 16S rRNA gene sequencing. Microbial community genomic DNA was extracted using the E.Z.N.A.® soil DNA Kit (Omega Bio-tek, Norcross, GA, U.S.) according to manufacturer's instructions. The hypervariable region V3-V4 of the bacterial 16S rRNA gene were amplified with primer pairs 338F(5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R(5′-GGACTACHVGGGTWTCTAAT-3′) by an ABI GeneAmp® 9700 PCR thermocycler (ABI, CA, USA), the hypervariable region V3-V5 of the archaea rRNA gene were amplified with primer pairs Arch344F(5′-ACGGGGYGCAGCAGGCGCGA-3′) and Arch915R(5′-GTGCTCCCCCGCCAATTCCT-3′). PCR amplification; agarose gelelectrophoresis; DNA purification and recovery; and DNA library sequencing preparation were all performed according manufacturer's instructions. Operational taxonomic units (OTUs) with 97% similarity cutoff were clustered using UPARSE version 7.1, and chimeric sequences were identified and removed. The taxonomy of each OTU representative sequence was analyzed by RDP Classifier version 2.2 against the 16S rRNA database (eg. Silva v138) using confidence threshold of 0.7.

The absolute content of 16s and ARCH archaeal genes in the samples was determined by using Real-time PCR assay, DNA samples prepared from the previous step were reacted in a 96-well plate in a fluorescence quantitative PCR instrument (Type ABI7300, Applied Biosystems, USA). The primers selected for the 16S rRNA gene were Eub338_Eub806, and those for the archaeal gene were Arch344F_Arch915R.

2.3.3. Metagenomic sequencing and analysis

Using E.Z.N.A.® Soil DNA Kit (Omega Bio-tek, Norcross, GA, U.S.) to extract total genomic DNA of the sample according to the manufacturer's instructions. Concentration and purity of extracted DNA was determined with TBS-380 and NanoDrop2000, respectively. DNA extract quality was checked on 1% agarose gel. The DNA was broken into fragments of approximately 400 bp using ultrasound (Covaris M220, USA). Sequencing library was constructed using the NEXTflex™

Rapid DNA-Seq. Subsequently, Majorbio Bio-Pharm Technology Co., Ltd. (Shanghai, China) used the Illumina NovaSeq/Hiseq Xten(Illumina Inc., San Diego, CA, USA) for sequencing. The raw reads from metagenome sequencing were processed on the free online platform of Majorbio Cloud Platform (cloud.majorbio.com).

2.4. Statistical analysis

An analysis of variance (ANOVA) was employed to evaluate the significance of results, and if the calculated p was less than 0.05, the corresponding result was considered statistically significantly. In this study, statistical analysis was conducted by IBM SPSS statistic.

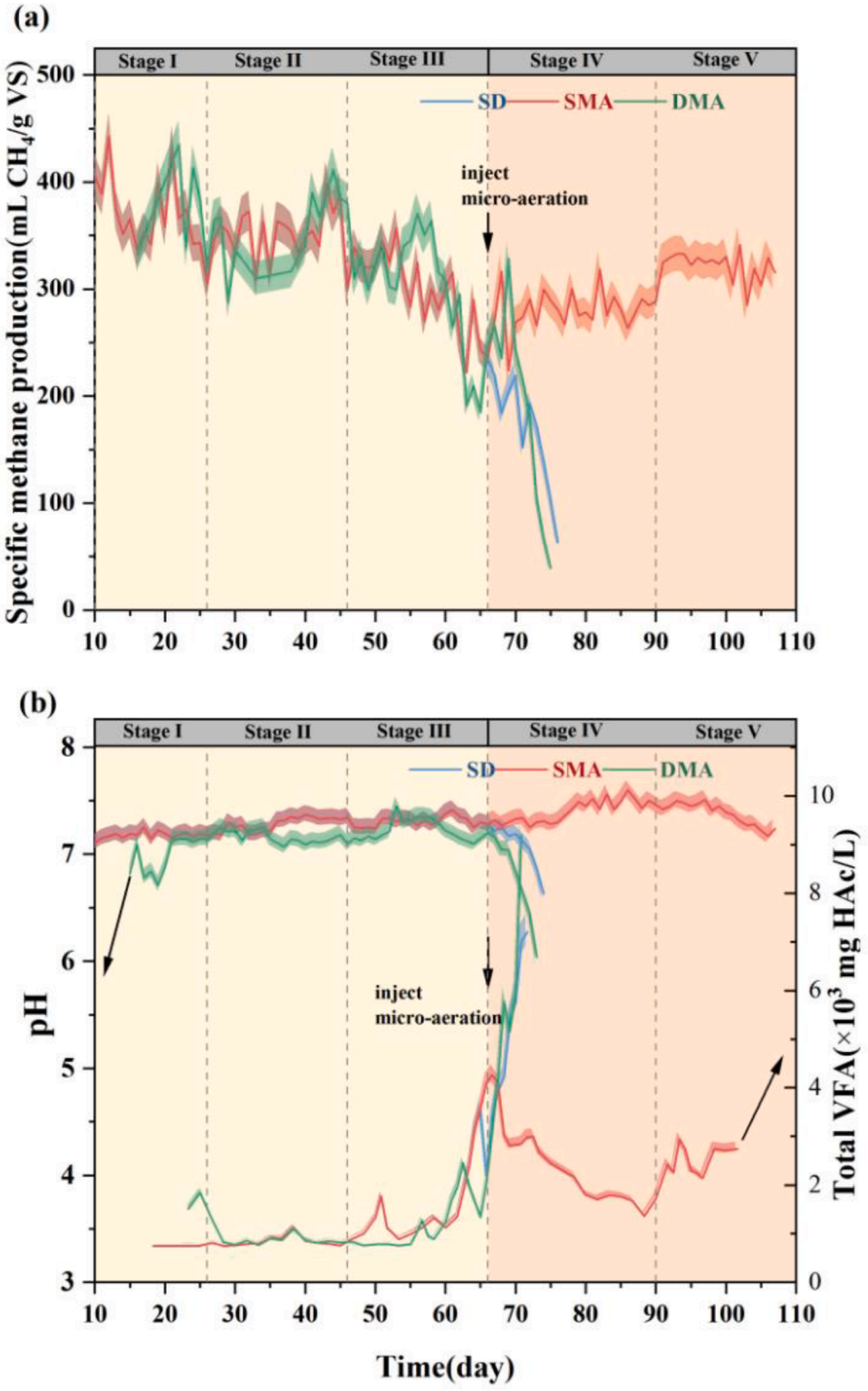


Fig. 1. Digestion performance in different stages.
((a) Specific methane production; (b) Total VFA and pH (During the first 66 days of the experiment, SD, SMA and DMA were operated parallelly)). The transparent part is the error band.

3. Results and discussion

3.1. Digestion performance

Fig. 1 shows the overall digestion performance with increasing OLR from 2.5 to 5.5 g VS/L·d and the effects of micro-aeration. Digesters run stable when OLR was maintained lower than 4.0 gVS/L·d (Stage II). When OLR was further increased to 5.0 gVS/L·d (Stage III), acid accumulation and subsequent acid inhibition occurred gradually due to overload of feeding organics. There is a significant decline in the specific methane yield from about 367 mL/gVS at the end of stage II to 339 mL/gVS after several days' digestion. By the end of stage III, the specific methane production was reduced by 33.07% relative to the end of stage II. After 20 days' operation under such OLRs, micro-aeration was introduced in the direct (DMA) and side-stream (SMA) modes, and a sharp contrast in the digestion performance was observed. There was intense acidification in the SD digester with significant decreases of methane production (Fig. 1a) and pH values. The VFA concentration climbed rapidly (Fig. 1b), while the ratio of VFA/ALK exceeded the warning value of 0.3 (Fig. S3), indicating the balance between acidity and alkalinity had been eliminated. Direct injection of air into the main digester (DMA) further accelerated the acidification, and a steep climb in TVFA concentration was observed. These two AD systems (SD and DMA) collapsed rapidly. The results for direct air introduction differ from previous studies reported by Nguyen et al. (2019) where direct micro-aeration restored the acidification in the digester. These differences may be due to distinct properties of feeding stock substrates. Compared to lignocellulosic biomass used by Nguyen et al. (2019), FW in this study presents much higher biodegradability, with much higher rates of hydrolytic-acidification. Therefore, direct air introduction further accelerated acidity accumulation. However, in this study, SMA responded opposite to DMA. The AD system showed an immediate response and side-stream micro-aeration prevented further decline in specific methane production (Fig. 1a), and the methane production properties were restored immediately to previous levels with increasing CH_4 content up to 60.01% (Table 1). Meanwhile, the TVFA concentration declined gradually restoring the pH of the slurry (Fig. 1b). After 20 days of the SMA, TVFA concentration decreased from 4.59 to 1.32 g HAc/L (Table 1). The lowered TVFA concentration assured the index of VFA/ALK ratio was maintained lower than 0.3 (Fig. S3), even at high OLR, which guarantees a good environment for methanogens. Also, propionate concentration decreased rapidly, while acetate concentration showed a small increase (Fig. S4). This result can be attributed to the conversion of VFA, such as propionate. Guo et al. (2021) also reported that moderate micro-aeration is beneficial to VFA consumption. The introduction of SMA prevented the decline of TCOD removal, and TCOD removal for the SMA system was maintained at about 69.2% (Table 1). These results indicate that the SMA digester was restored to stability and can run efficient digestion of FW at high OLRs.

The above results have shown the potential of SMA to relieve the acidity accumulation under high OLR. Therefore, this study further increased the OLR from 5.0 to 5.5 gVS/L·d (stage V) to explore SMA limitations. Specific methane yield further improved to about 330 mL/g VS under such OLR, and the methane production performance was stable, even at the end of this stage. It is interesting that though total VFA concentration in this stage climbed gradually to higher than 3000 mg HAc/L with the slow decline of pH to below 7.2 (Fig. 1b), the system still ran properly. This reflects that SMA improves the tolerance of the system and optimizes the operational performance of anaerobic digestion in various aspects.

Changes to ORP in the SMA and DMA reactors were also recorded during a day in stage IV, and the results are shown in Figure S5. The application of side-stream micro-aeration brought a slight increase in ORP, and the ORP in slurry stabilized gradually to base levels. However, in the DMA reactor, a steeper climb and long duration time of the ORP was observed.

3.2. Enzymes activity

Micro-aeration in AD is believed to promote hydrolytic and fermenting microorganisms, so it is necessary to understand the positive effects of SMA on hydrolytic-acidification to reveal potential mechanisms. Fig. 2 compares enzyme activity before and after SMA treatment. α-glucosidase plays a role in the hydrolysis of polysaccharides (Ruan et al., 2019). Increased α-glucosidase activity after aeration was involved in the hydrolysis of polysaccharide-like organics. This supports the recovery of high levels of methane and organics conversion after systemic acid accumulation. Acetate kinase, a key enzyme participating in acetate production, increased significantly after aeration. Increased activity of α-glucosidase and acetate kinase verified the enhanced hydrolytic-acidification process brought by micro-aeration. In addition, the ATP content in the system increased significantly after aeration, indicating that micro-oxygen promoted the growth and metabolism of microorganisms resulting in more microbial activity and hydrolase production. For methanogenesis, the activity of F_{420} declined after treatment ($p<0.001$). F_{420} is commonly used as an electron donor in the reduction of CO_2 to CH_4, lowered F_{420} indicates hindered CO_2 reduction, and it is quite possible that the acetoclastic methanogenesis pathway may dominate after SMA treatment (Wunderer et al., 2022). The introduction of oxygen created reactive oxygen species (ROS), hence the level of CAT was observed a slight increase (Wu et al., 2021). However, CAT content remained low, likely due to other mechanisms in SMA, and the level of ROS in SMA may not be high.

Microbial bioactivity before and after SMA treatment was assessed

Table 1
Digestion performance for different groups in different stages.

	SMA						SD		DMA	
	Stage I	Stage II	Stage III unstable		Stage IV	Stage V	Stage IV unstable		Stage IV unstable	
			Initial	Final			Initial	Final	Initial	Final
OLR, gVS/L·d	2.5	4	5	5	5	5.5	5	5	5	5
pH	7.18	7.28	7.35	7.25	7.37	7.25	7.22	6.63	7.20	6.06
Total VFA, mg HAc/L	72.5	364.5	807.5	4598	1320	2382.1	4356.9	8819.7	4690.3	11528.9
Specific methane production,mL CH_4/g VS	356.5	366.9	338.9	245.6	287.8	322.1	234.8	63.3	265.4	39.4
Total alkalinity, g/L as $CaCO_3$	4.16	6.27	6.31	8.62	10.23	10.98	8.21	7.51	8.37	7.29
CH_4 content, %	66.22	59.02	57.98	54.95	60.01	59.84	54.91	42.99	53.92	41.64
VFA/ALK ratio	0.01	0.05	0.13	0.53	0.13	0.22	0.53	1.17	0.56	1.58
Acetate, g/L	0.04	0.09	0.31	1.77	0.33	0.52	1.85	3.41	2.18	2.97
Propionic acid, g/L	0.02	0.12	0.31	1.42	0.56	0.68	1.46	3.25	1.57	3.83
Isobutyric acid, g/L	0.00	0.05	0.11	0.53	0.33	0.38	0.31	0.64	0.26	0.83
Butyric acid, g/L	0.00	0.01	0.01	0.39	0.02	0.14	0.31	0.55	0.37	0.90
VS removal, %	77.08	77.06	77.06	79.84	73.75	73.29	78.71	72.64	78.25	71.33
TCOD reduction, %	69.89	69.02	71.89	68.32	69.18	68.7	67.52	64.56	68.76	60.55

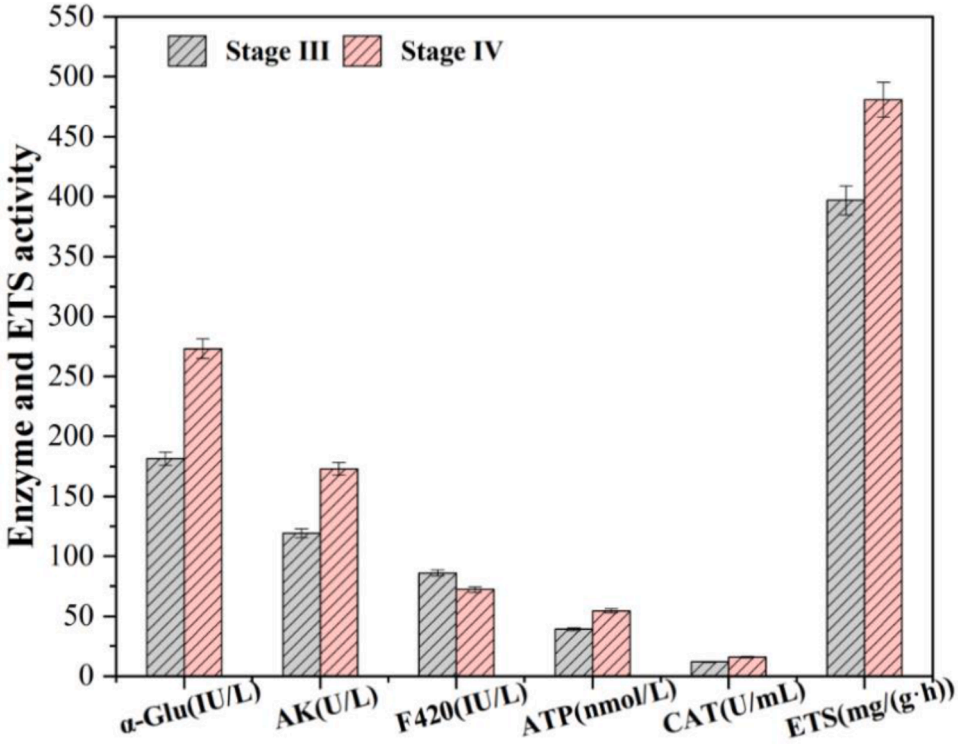


Fig. 2. Enzyme activity and ETS before and after SMA treatment.

by monitoring electron transport systems (ETS) during microbial metabolism (Fig. 2). Electron transport usually occurs in the cytoplasmic membranes of bacteria that generate flavoproteins by coupling the oxidation of NADH and other substrates to the protons. In aerobic respiration, quinone acts as a hydrogen carrier to transfer electrons to appropriate terminal electron acceptors such as O_2 producing cytochromes. O_2 can easily act as an electron acceptor in ETS, thus increasing activity (Wang et al., 2020). After SMA treatment, ETS activity in the digester increased significantly by 21%. This increase probably stemmed from the fact that trace amounts of oxygen added by SMA became the optimal electron acceptor. Notably, cytochromes generated by electron transport are important for overcoming oxidative stress, which is another way SMA may tolerate ROS. In addition, enhanced ETS activity also facilitates metabolism by maintaining redox homeostasis, which may contribute to digester stability.

3.3. Carbon mass balance

Samples were taken before and after SMA treatment to build carbon mass balance (Fig. 3). The carbon contained in the feeding stock was dominated by carbohydrates and proteins, which are partly converted to biogas and partly used by cells for synthesis. Carbon recovery in the form of CH_4 can be used to indicate the return of system stability. CH_4 increased from 18.8% in the unstable period (stage III) to 22.0% in stage IV after the application of SMA, and methane content in the biogas was significantly higher (Table 1). It is worth noting that the system faced severe pressure due to high OLR during this period. This finding further supports the important role of SMA in maintaining systematic stability. Moreover, in stage IV, the proportion of carbon recovered in the form of intermediates decreased significantly, while the carbon recovered in the form of digestates increased significantly. The decrease of intermediates may be ascribed to the enhanced conversion of particulate matter into soluble compounds, resulting in the increased carbon content in the digestates. The enhanced conversion of particulate organics indicates metabolic activity of fermentating bacteria in response to SMA in these stages. Combining the results observed in section 3.1 and 3.2, we inferred that the introduction of SMA restores the system from collapse by converting or consuming SCFA. Additionally, SMA promoted microorganisms metabolic activity, and thus, the hydrolytic-acidification processes were improved providing sufficient intermediates with low molecular weight for subsequent methanogenesis. Although the carbon balance could not clarify the specific path of carbon flow to methane, the conversion of VFA and the changes of key enzymes indicate changes in methanogenesis in response to SMA.

3.4. Microbial analysis

3.4.1. CLSM

In the CLSM image, SYTO 63 stains nucleic acids, Con A-AF555 (Fluorescently labeled lectins Concanavalin A) stains the α-mannopyranosyl and α-glucopyranosyl sugar, FITC (fluorescein isothiocyanate) stains proteins and other amine-containing compounds such as aminosugars in the granules, calcofluor white stains β-D-glucopyranose polysaccharides, Merged represents the combined image of four dyes.

Fig. 4 shows the distribution of components in the sludge observed in CLSM. According to the fluorescence intensity, a large number of filamentous microorganisms were observed in both samples, and the cells in the samples after SMA treatment were more densely distributed tending to form tightly aggregated morphology (Fig. 4). In addition, after SMA treatment, both polysaccharides (including α-mannose, α-glucose, and β-D-glucopyranose polysaccharides) almost cover the area where cells were located, while protein only appears in some areas. We speculate that the sample forms a tight core, and the glucose polysaccharide aggregates at the outer layer of the cell granules, similar to McSwain (2005). This may be related to EPS, which is capable of aggregating microorganisms. EPS with polysaccharide, as a skeleton, agglomerates microorganisms and organics. This form of a compact whole is favorable for electron transfer and metabolic processes. In addition, this form may resist the outside adverse environment. It can gather obligate anaerobes in the inner space and eliminate the toxic effects of ROS (singlet oxygen ($^1O_2$), superoxide anion radicals ($O2\cdot^-$), hydrogen peroxide (H_2O_2) and hydroxyl radicals (OH·)) (Nguyen and Khanal, 2018). Meantime, the

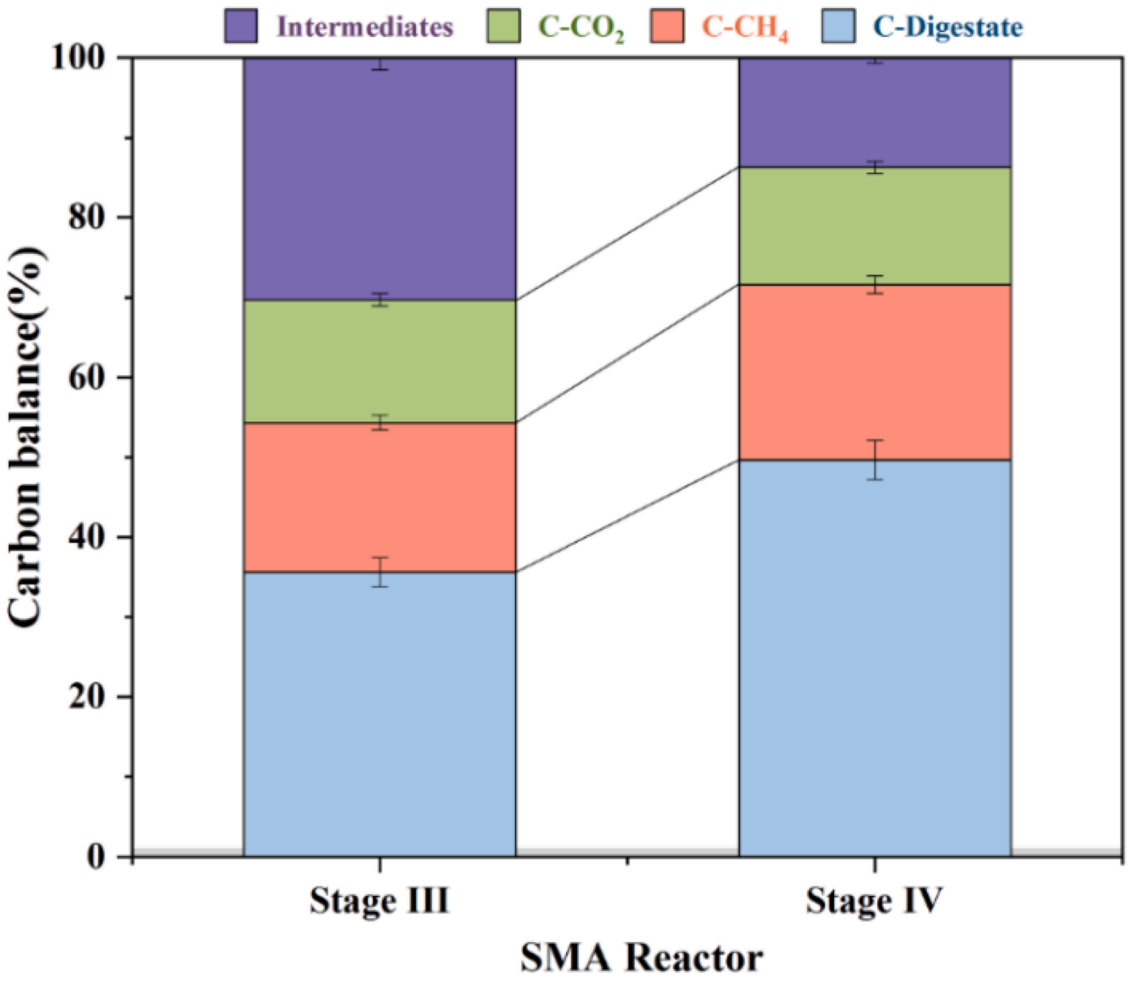


Fig. 3. Carbon mass balance in stages III, and IV.

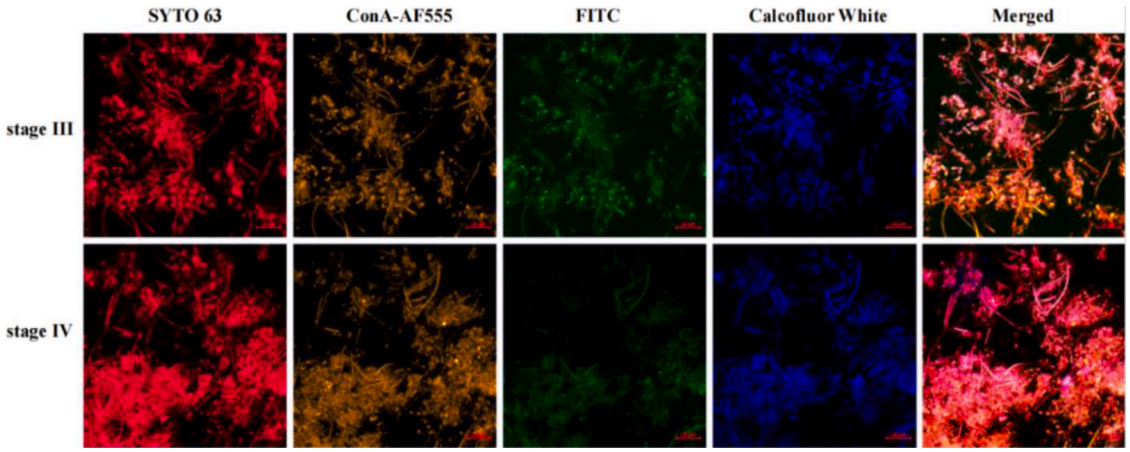


Fig. 4. CLSM diagrams of sludge before and after SMA treatment.

attached antioxidant enzymes and polyphosphate can help cells to tolerate extreme environments.

3.4.2. *Diversity*

Samples were taken at the end of each stage for microbial analysis. Inoculum sludge was also taken for comparison. Indices related to microbial richness and diversity were analyzed. In general, the level of species evenness was proportional to microbial diversity. Higher values of Sobs, Ace and Chao indicate higher microbial richness. Shannon and Simpson indices are negatively correlated, while a higher Shannon index indicates higher microbial uniformity (Xu et al., 2021). In the bacterial community, the introduction of SMA was found to resulted in significant decreases in both species richness and uniformity (Table 2), implying that SMA reduced bacterial diversity. The phenomenon in here may be due to that trace oxygen stimulate the growth of facultative bacteria and limit the growth of bacteria sensitive to oxygen. Meanwhile, there is also

Table 2
Microbial diversity for inoculum sludge and sludges before and after SMA treatment.

Items	Groups	Community richness			Community diversity		Community coverage(%)
		Ace	Chao	Sobs	Shannon	Simpson	
Bacteria	RAW	1222.40	1214.88	1205	5.58	0.01	99.90
	Stage III	469.37	513.28	373	3.03	0.14	99.76
	Stage IV	336.38	337.80	282	2.99	0.12	99.85
Archaea	RAW	179.84	174.00	160	3.12	0.11	99.93
	Stage III	197.83	193.55	184	3.00	0.14	99.92
	Stage IV	139.13	168.20	113	1.86	0.41	99.91

a huge impact on the metabolic pathways, micro-aeration in side-stream reactor may create an environment super suitable for the growth of dominant bacteria, so that the ecological niches in the digestive system are gradually occupied by them, resulting in the decrease of the diversity. In addition, the bacterial diversity in the inoculum was significantly higher than in other samples, which can be attributed to the efficient digestive environment in the reactor for treating sludge. Similarly, archaea had lower microbial diversity in the SMA sample, revealing that the archaea community was purified by the SMA. This kind of community may pose much higher activity in converting intermediates (acetate and H_2) into methane.

3.4.3. *Microbial community structure*

In order to reveal the effect of introducing of SMA on the microbial community structure, this work examined the microbial community compositions for sludges before and after SMA treatment. Meanwhile, q-PCR was used to understand the variation of absolute gene numbers in the SMA treatment (Fig. 5). There were significant increases in the copy number of target genes in bacteria and archaea after SMA, showing that SMA treatment accelerates microbial proliferation. The rate of microbial reproduction depends on cell growth and division, which generates a strong demand for energy and biosynthesis, since cell division-dependent proteases (such as SulA) require ATP to function. Because ATP is synthesized only when needed due to thermodynamic constraints, the observed increase in ATP content also attests to the activity of microbial growth and reproduction (Fig. 2).

Specific to the composition of the microbial community (Fig. S6a), most sequences in SMA were assigned to seven phyla: Firmicutes, Actinobacteriota, Bacteroidota, Spirochaetota, Synergistota, Elusimicrobiota, Chloroflexi. Except for Bacteroidota and Spirochaetota, the relative abundance of other bacterial phyla was significantly increased by SMA, suggesting that these microorganisms are more adapted to the environment created by SMA. In fact, the first three dominant bacterial phyla in SMA are all reported to be facultative bacteria, often found in micro-aerobic fermentation systems (Lim et al., 2014).

Among them, Firmicutes became the dominant bacteria after SMA treatment. It has been reported that Firmicutes are important hydrolytic-acidifying bacteria with resistance to high temperature and low pH (Pervin et al., 2013). Its high abundance explains the higher efficiency in organics hydrolysis and subsequent acidification. Actinobacteriota were the second dominant bacteria that play an important role in degradation of organic matter and forming humus (Zhang et al., 2012). They favor sludge aggregation and tend to form stable coated agglomeration particle structures, which explain the changes in microbial morphology observed in Section 3.3.1. Synergistota are able to accelerate the conversion of VFA to acetate (Rivière et al., 2009). Their significantly increased abundance is thought to be the key to resolving excessive acidity (Ruan et al., 2019). The Synergistota in SMA accelerates the consumption of VFA by converting them to acetate that can be further used by methanogens. The improved absolute and relative abundance of the above bacteria in SMA implies a cooperation among hydrolysis, acidification, and subsequent acetate production in AD. Elusimicrobiota contained a single, facultative species with a cluster of oxygen stress protection genes (Guo et al., 2021), and its proliferation may explain the increase in CAT content. Both Chloroflexi and Actinobacteriota can degrade complex substrates by producing hydrolysis enzymes (Ziganshin et al., 2013). Bacteroidota, which contain hydrolytic and acid-producing bacteria involved in the hydrolysis of complex polysaccharides (Guo et al., 2014), was slightly reduced after SMA treatment. Spirochaetota are obligate anaerobes whose relative abundance

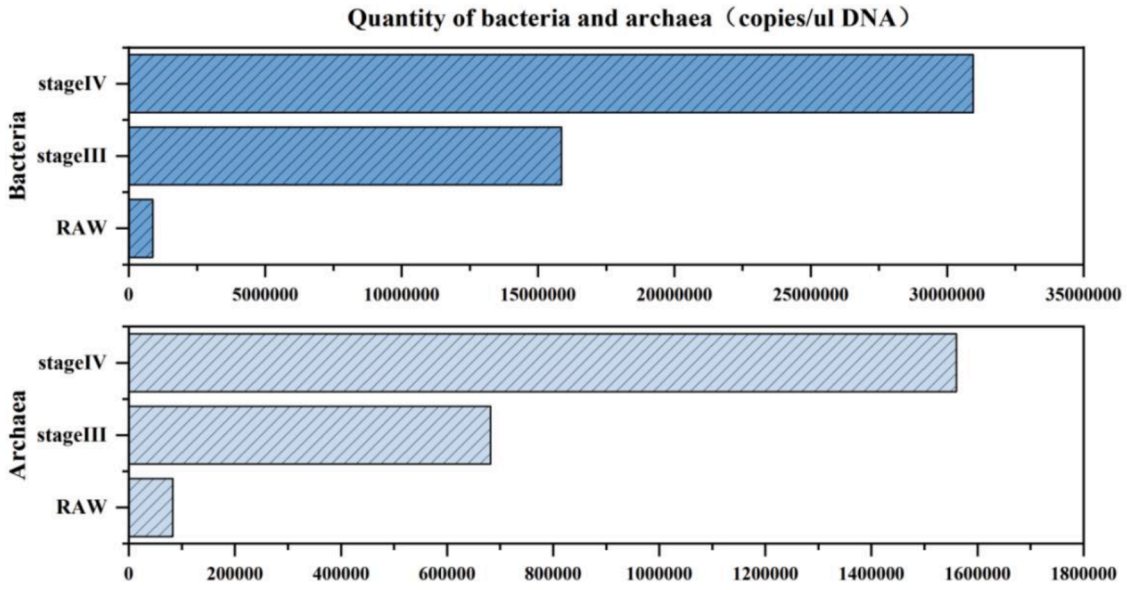


Fig. 5. Absolute abundance of bacteria and archaea in SMA reactors and inoculum sludge.

and total copy number decreased significantly under the micro-aerobic condition with SMA (Huang et al., 2021).

At the genera level, the community structure of archaea was much simpler than the bacterial community. Methanogenic archaea dominated in all samples (Fig. S6b). For the SMA treated sample, the main genera were *Methanosaeta, Methanolinea, Methanofastidiosum, and Methanothermobacter*. After long acclimation under high OLR feeding of FW (Stage III), the archaea community structure reshaped. The relative abundance of *Methanosaeta*, and some nonranked Archaea increased. The introduction of SMA changed the community structure dramatically. Under the effect of SMA, *Methanosaeta* proliferated rapidly and occupied almost all ecological niches, indicating that SMA has a

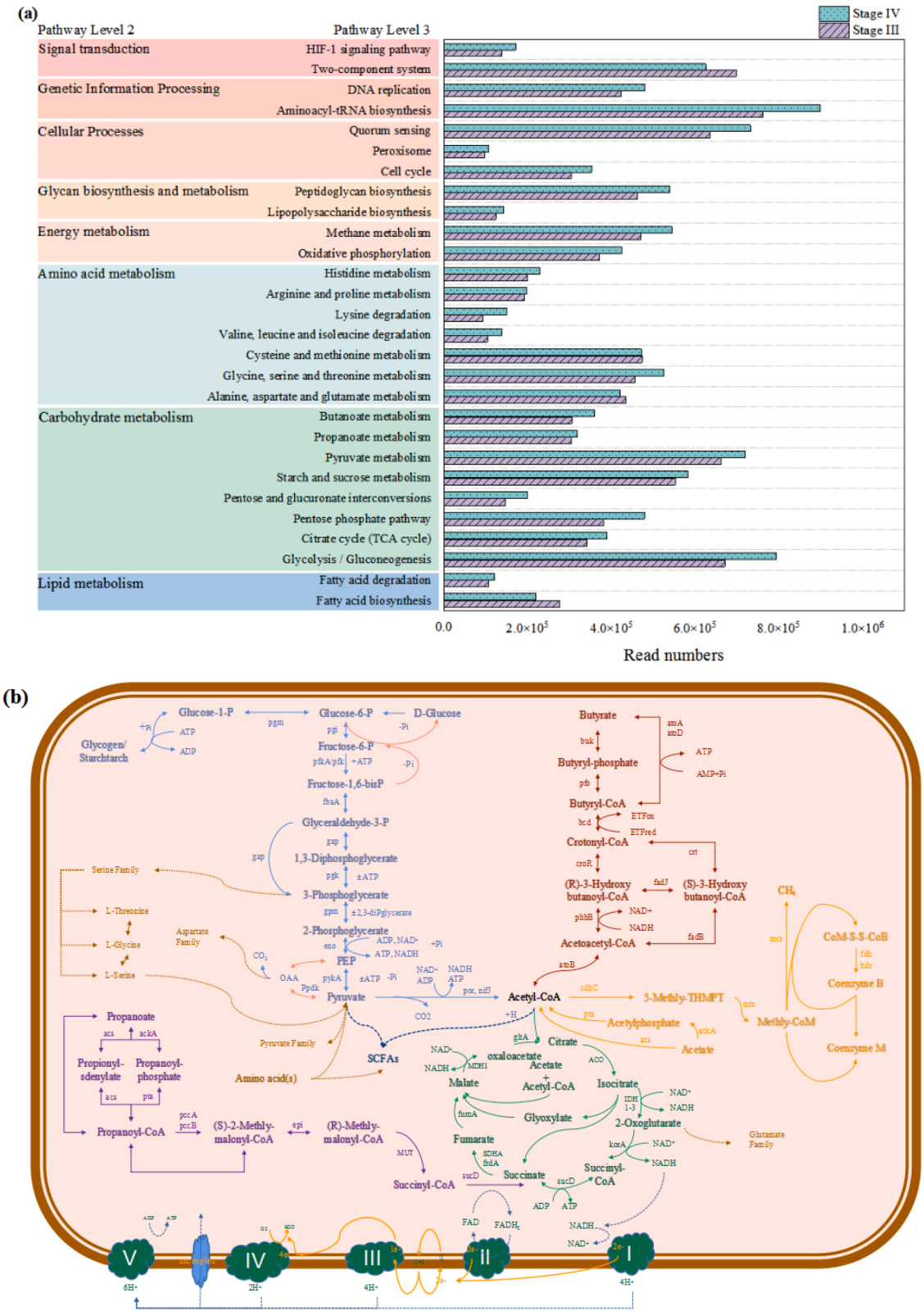


Fig. 6. (a)KEGG pathway and (b)metabolic process reconstruction.

domestication effect. *Methanosaeta* is an obligate acetatoclastic methanogen that produces CH_4 by direct acetate decomposition. The highest abundance in the SMA participated stage indicates that aceticlastic methanogenesis is its main methanogenesis pathway, this is also supported the enzyme results. In addition, other enriched archaea found in the SMA reactors were *Methanomicrobiales, Methanobacterium*, and *Methanoculleus. Methanobacterium* is highly associated with hydrotropic methanogenesis and can enhance interspecies H_2 transfer during AD (Zheng et al., 2021), *Methanoculleus* are hydrogenotrophic methanogens, which are capable of producing CH_4 by reducing CO_2 with H_2 or formate as an electron donor (Dong et al., 2022). The low relative abundance of these hydrogenotrophic methanogens explains why H_2 was barely detected throughout the digestive process. In AD of perishable organics under high OLR, with the fast formation of VFA, quantities of electrons were released and combined with protons to produce H_2, which needs hydrogenotrophic methanogens to be consumed to ensure syntrophic oxidation to convert propionic and butyric acid into acetate. SMA reduced H_2 generation by accelerating its consumption by introducing O_2. The lower hydrogen partial pressure eliminated the thermodynamic barrier in syntrophic oxidation, resulting in fast conversion of accumulated VFA into acetate, thus inducing acetoclastic methanogens.

Methanogenesis in SMA differs from previous direct micro-aeration. According to previous studies, oxygen directly stimulates facultative anaerobes, and the micro-aerobic environment screens out methanogens intolerant to oxygen. Due to higher tolerance of hydrotropic methanogens such as *Methanobacterium* and *Methanoculleus*, their abundance was higher under direct micro-aeration. However, in this study, ROS in SMA were weakened in the cycle, which favors the role of *Methanosaeta. Methanosaeta* is sensitive to oxygen but has an advantage in metabolism, similar to its higher abundance in reactors with low oxygen (Guo et al., 2021). In addition, intermediates, energy and ATP still play a role, which gives *Methanosaeta* a more significant advantage in SMA. This is the reason for the formation of the unique methanogenic pathway in SMA.

3.5. Metagenomic analysis

In order to identify the potential mechanisms for the heightened performance of high-load AD and the reduction of acid inhibition under SMA, this study sampled the genes involved in the metabolic processes at the end of stages III and IV to analyze them using macrogenomics, while annotating genes and key KEGG pathways (Fig. 6a). Metabolic pathways were reconstructed manually in order to gain a comprehensive understanding of metabolic processes (Fig. 6b).

3.5.1. Conversion and utilization of organic compounds

The KEGG pathways were significantly different under SMA (Fig. 6a and Table S3). Most of the key KEGG pathways detected in the samples were up-regulated, which is consistent with enhanced anaerobic digestion performance (Fig. 1a). Pathways associated with carbohydrate metabolism showed a significant increases, which explains rapid VFA depletion (Fig. 1b) and the subsequent gradual recovery to systematic stability.

The Embden-Meyerhof-Parnas pathway (EMP) is the main pathway of glucose catabolism, where pyruvate is the final product. The expression of key genes in this process is shown in Table S4. The expression of *pfkA/pfk* was the highest, using ATP to phosphorylate F-6-P to F-6-bP (FBP, Fructose bisphosphate), which belongs to the glycolytic pathway. In addition, *pfk* is a suppressor gene of gluconeogenesis, and another suppressor gene *pyk* was also highly expressed after SMA treatment. *pgk* was most significantly upregulated after SMA treatment. It is involved in the conversion of 1,3-Diphosphoglycerate and the generation of ATP from ADP. Its activation was reported to be dependent on the catabolic control protein (CcpA, K02529), a global regulator that inhibits multiple catabolic manipulators involved in the secondary metabolism of carbon (Xiao et al., 2021). *Ppdk* was also highly expressed after SMA treatment, and it is involved in the interconversion of pyruvate and phosphoenolpyruvate (PEP). Since one glucose can be converted into two PEP molecules in the EMP pathway, whereas *pyk* can only convert one PEP to one pyruvate, a sudden increase in the upstream reaction activates *ppdk* which is then responsible for catalyzing the reversible conversion of PEP, AMP and PPi to pyruvate, ATP and P (Cui et al., 2020). The expression of these genes demonstrate that SMA optimizes the balance of interconnected nodes in organic transformation and enhances the efficiency of favorable transformation processes.

The pentose phosphate pathway (PPP) is another glucose oxidizing pathway, and some of its products can re-enter the EMP. PPP was found to play an important role in maintaining systematic stability after SMA treatment. Non-oxidative PPP synthesizes key precursors such as amino acids, nucleotides, and co-factors, and contributes to the maintenance of glycolytic fluxes (Rashida, 2021). PPP was significantly enhanced in SMA (Table S4). In addition, non-oxidative PPP can direct the products of oxidative PPP (R5P) into glycolysis, which helps maintain NADPH levels. This result verifies that SMA stimulates the interconnection between PPP and glycolysis. The reversibility of non-oxidative PPP helps the system achieve dynamic balance via intermediates.

Amino acid metabolism is coupled with PPP, which is also shown to be enhanced by SMA. Glycine, serine and threonine metabolism pathways were significantly enhanced (Fig. 6a). The serine family is synthesized via 3-phosphoglycerate and regulated by it, and the metabolism of the serine family can promote EMP. The serine family is usually converted to pyruvate by deamination and can contribute to the TCA cycle through pyruvate and acetyl-CoA (Ye et al., 2018), which is important for organic matter conversion and energy metabolism.

The above results suggest that SMA promotes both carbohydrate and amino acid metabolism. The enhanced conversion and utilization of organics induced by SMA allows cells to produce sufficient ATP/energy, accumulate biomass, and maintain growth, which enhances the ability of microorganisms to cope with environmental stress.

3.5.2. SCFA conversion

Different fermentation processes have different abilities to oxidize NADH and produce H_2 and ATP. To avoid the accumulation of large amounts of NADH from oxidative dehydrogenation of substrates and to maintain the $NADH/NAD^+$ balance, the production of acetate is usually accompanied by other products (e.g propionic acid, butyric acid or ethanol). The conversion of acetyl-CoA to butyric acid has been reported to consume two molecules of NADH (Cheng et al., 2020). Therefore, fast conversion of acetyl-CoA avoids the accumulation of NADH that would enhance the accumulation of FAs, especially in high OLR AD. SMA showed a rapid response to alleviate SCFA accumulation with a significant upregulation of the fatty acid oxidation pathway (Fig. 6a). The involved enzymes are presented in Table S5.

This work constructed a complete β-oxidation pathway for butyric acid metabolism. The most pronounced upregulation of *buk* (1.70-fold promotion) activates butyrate to butyryl-CoA, and another enzyme involved in this process, *atoD*, was also actively expressed (1.67-fold promotion) with one ATP synthesized per butyrate molecule in this step (Hao et al., 2020). Most *atoD* originates from Firmicutes, confirming their increased abundance (Fig. S6a). *AtoB* is the most strongly expressed enzyme in SMA and is responsible for the final step in producing acetyl-CoA. The upregulation of *atoB* contributes to the distribution of precursors in the acetyl-CoA branch node and is essential for the conversion of butyrate to acetate (Kim et al., 2018). In addition, the greatest downregulation was observed in *fadR*, which is involved in the oxidation to acetoacetyl-CoA via (S)-3-Hydroxybutanoyl-CoA. *fadR* is a multifunctional regulatory gene that negatively regulates fatty acid degradation by reducing the transcription of the *fad* structural gene (Sun et al., 2020). Its downregulation further verifies that fatty acid degradation was enhanced in SMA.

Propionate oxidation via the conventional methylmalonyl-CoA

pathway where enzymes (*acs, ackA, pta*) convert to propanoyl-CoA was in much higher abundance than other enzymes. SMA strongly catalyzed propionyl-CoA synthesis, and the product of propionyl-CoA carboxylation could release ATP and reduce the equivalent in the succinyl-CoA synthesis step (Hao et al., 2020). In addition, the expression of *pccA* was significantly upregulated (2.14-fold promotion), and its main function was propionyl-CoA carboxylation, accompanied by CO_2 formation. The upregulation of *pccA* avoids the accumulation of propionyl-CoA, where intracellular accumulation of propionyl-CoA inhibits mitochondrial metabolism and reduces the synthesis of citrate and ATP (Wongkittichote et al., 2017). *sucD* was another highly expressed enzyme involved in the conversion of succinyl-CoA, which is an intermediate in the TCA cycle. The upregulation of *sucD* is due to the activation of TCA under micro-aerobic conditions, which makes it difficult to accumulate succinyl-CoA.

Furthermore, the enhanced conversion of propionic and butyric acid reflects enhanced syntrophic oxidation to the production of acetate. Pyruvate is an important node in acetate conversion (Table S5). The gene for the conversion of pyruvate to acetyl-CoA (K03737) also showed significant upregulation, while *por* and *nifJ* were the most abundant enzymes in all relevant metabolic processes. The process of transforming pyruvate to acetyl-CoA is essential for metabolism and energy production. In addition, electrons (e^-) released from oxidized pyruvate are transferred via flavodoxin to various reductases that overcome oxidative stress and restore the reducing environment in the cell (Nakayama et al., 2013).

3.5.3. Methanogenesis

Both the microbial community analysis (Fig. S6b) and the efficient conversion of VFA to acetate suggest that acetoclastic methanogenesis dominates methane production. According to the KEGG database, three methanogenic pathways exist, including hydrogen, acetate and methyl nutrient pathways (Table S6). Fig. S7 shows the genes encoding the acetate pathway dominating after SMA treatment with very low expression of the hydrogen and methylotrophic pathways. This result is consistent with the results of related enzymes (Fig. 2).

In the acetoclastic pathway, acetate can be converted to acetyl-CoA via two ways. The enzymes involved in both processes were highly expressed. The abundance of the coding gene for *acs* was high; it has been reported that among C2-C5 short chain fatty acids, *acs* preferentially utilized acetate with similar affinity (Fujino et al., 2001). The activation of this enzyme may be responsible for the increased acetate utilization. Also, *Methanosaeta* has been shown to use acetyl-CoA synthetase to convert acetate (Smith and Ingram-Smith, 2007), which explains the proliferation of *Methanosaeta* (Fig. 5).

The hydrotropic methanogenesis is an important process in previous micro-aeration studies, but it was not highly expressed in SMA. The small amount of hydrogen methanogenesis in SMA is responsible for converting excess H produced by metabolism. The most highly expressed enzymes in this pathway, *fmd*, and specifically *fmdE*, were reported to be closely associated with *Methanobacterium* (Zheng et al., 2021), which is consistent with CO_2-using methanogens that was found in the archaeal community structure (Fig. S6b). The results further verifies the differences between DMA and SMA in terms of the archaea community and metabolism.

3.5.4. Energy metabolism

Due to the introduction of O_2 into the anaerobic environment, cells should first use O_2 as an electron acceptor because it is more favorable in thermodynamics, and microorganisms are able to adapt their metabolism pathways from anaerobic to micro-aerobic. The transition of the metabolism is accompanied by the changes in energy acquisition. Energy metabolism exhibits a more aerobic-like structure with increased levels of oxidative phosphorylation, TCA, and PPP (Table S7).

Under micro-aerobic conditions, acetyl-CoA undergoes the TCA cycle and is further oxidized to CO_2 using the electron carrier NADH. The proton flow in NADH conversion drives ATP production through oxidative phosphorylation to support microorganism growth metabolism. Pyruvate entering the TCA cycle under micro-aerobic conditions provides eight times more ATP than that produced by substrate-level phosphorylation under anaerobic conditions (Nguyen and Khanal, 2018). In this study, SMA stimulated key enzymes in the TCA cycle including *gltA, IDH1-3* and *korA*. Among them, *gltA* is the first step in the entry of acetyl-CoA into the TCA cycle and competes with other transformations of acetyl-CoA (Tovilla-Coutiño et al., 2020). The upregulation of *gltA* and *IDH1-3*, and the increased expression of *korA* during TCA. This suggests that SMA has potential for microorganisms to undergo a complete TCA cycle providing more energetic support for the proliferation of microorganisms.

Oxidative phosphorylation is another important source of energy for the cell. When H^+ from carbohydrate metabolism is transferred to the hydrogen acceptor NAD via a membrane-associated proton-translocating ATPase, energy containing proton motive force (Δp) can be used to generate ATP (Nolfi-Donegan et al., 2020). Significant expression or upregulation of quinone oxidoreductase (M00144), succinate dehydrogenase (M00149), fumarate reductase (M00150) and cytochrome c oxidase (M00154) in SMA is the main manifestation of the enhanced oxidative phosphorylation. However, electron transport-dependent complexes I, III and IV are sources of ROS. However, under high NADH/NAD^+ conditions, $O_2\cdot^-$ production is positively upregulated (Murphy, 2009). Thus, SMA promotes microbial growth and metabolism in favor of avoiding NADH accumulation, which would reduce ROS stress.

In addition, *ppx*, a key enzyme in the metabolism of the high energy storage substance polyphosphate (*polyP*), was also observed to be highly expressed after SMA treatment. *ppx* releases the phosphate group at the end of the *polyP* chain which releases a large amount of energy, and its involvement also explains the high cellular yield after SMA treatment, demonstrating that the energy inherent in polyphosphate and PPi is efficiently utilized.

3.5.5. Oxidative stress regulation

The micro-aerobic environment created by SMA acts directly on the side-stream reactor. After slurry circulation, the ROS formed decays, and most microorganisms in the main reactor receive limited stimulation, resulting in oxidative stress regulation in the system.

First, in SMA, *polyP*, a macromolecular chain polymer and its associated enzymes, present an important regulatory role in facing oxidative stress. Under extreme conditions, the cellular self-protection mechanism centered on *polyP* is mobilized to promote the formation of a biofilm as a protective barrier against hydrogen peroxide (Tunpiboonsak et al., 2010). This explains the upregulation of lipopolysaccharide and peptidoglycan-related gene expression (Fig. 6a). In addition, *polyP* acts as a protein chaperone and resists oxidative stress by preventing protein denaturation through translated polyphosphorylation, which occurs by a non-enzymatic mechanism (Gray et al., 2014). The identification of *PPK* and other enzymes involved in *polyP* metabolism in SMA also demonstrates that *polyP* facilitates oxidative stress regulation. In addition, the increase in CAT (Fig. 2) was evidenced by the upregulation of K03386 (peroxiredoxin). The high expression of K04564 (superoxide dismutase) also demonstrates the regulatory process of oxidative stress. SodA and SodB protect DNA and proteins from oxidative damage by ROS (Wu et al., 2021). A two-component regulatory system controls *cyo* and *cyd*. The expression of *cyd* was higher in SMA (1.86-fold) because *cyd* has high affinity and is more adapted to low oxygen concentration. In addition, anaerobic bacteria can use NADH oxidase to eliminate the effects of oxygen, and one of the enzymes observed in SMA, *fprA1_2*, can catalyze the reduction of oxygen to H_2O (Hillmann et al., 2009). Upregulation of the superoxide reductase system, which occurs in anaerobic bacteria, was also observed. Superoxide reductase *dfx* reduces superoxide to hydrogen peroxide without forming molecular oxygen. This system has a coordinated role with NADH oxidase (Jenney et al., 1999).

The expression of oxidative stress-related genes was upregulated in SMA, but their expression was low relative to other genes involved in metabolism. The antioxidant function performed by anaerobic bacteria still occupied a large proportion. This phenomenon supports the hypothesis that a microbial community dominated by facultative anaerobes was formed in the side-stream reactor to overcome ROS. More importantly, O_2 was fully utilized in the side-stream reactor to promote energy metabolism to produce associated products. These products were exchanged within the system through circulation. The main reactor forms a microbial community dominated by anaerobes, complementing NADH oxidase to scavenge the remaining ROS and using the favorable substances provided by the side-stream reactor. In other words, SMA treatment simultaneously limited the negative impacts and improved the ability of microorganisms to cope with stress, which may also be the main reason why SMA is superior to SD and DMA.

3.6. Overall summary

SMA ensured stable performance of the digester under high OLR, with more prominent acid-relieving inhibition and better tolerance to ROS. Environmental stress brought by SMA profoundly reshaped the microbial community structure, causing a dramatic shift in metabolic pathways. At the same time, enhanced metabolism maintained cell proliferation, which facilitated rapid adaptation to high OLR and oxidative stress, eventually restoring system stability (Fig. 7).

Specifically, SMA treatment enhanced the dominant species of hydrolytic acidifying bacteria such as *Firmicutes, Actinobacteriota* and *Methanosaeta*, and enhanced hydrolysis, fermentation, and methanogenesis. First, SMA enhanced the conversion and utilization of organic matter by promoting EMP, PPP, and amino acid metabolism, and enhanced electron shunting from SCFA to acetate. In addition, trace oxygen provided by SMA optimized the pathway of energy metabolism, and the slurry circulation operation fully exchanged metabolic intermediates, related key enzymes, and ATP to meet the material and energy requirements for cell proliferation. Meanwhile, SMA effectively mitigates the detriment of ROS and facilitates the stable operation of the system.

In conclusion, SMA-based mesophilic anaerobic digestion is a promising strategy for high-load, easily hydrolyzed substrates. Because SMA can withstand higher OLR, it can efficiently utilize resources and reduce the building footprint, generating greater economic benefits. The process of SMA inherits the advantage of conventional micro-aeration, while the structure avoids the negative effects of ROS and maximizes the benefits of micro-aeration, which helps large-scale promotion. However, there are still many problems need to be solved in scaling up. System controlling would be the first problem. Comprehensive computer program control system needs to be developed to control the transportation of slurry from each other, as well as the intensity and duration of micro-aeration, by referring the results of online ORP, water level, and air flow rate monitoring. Additionally, relatively strong mixing is required in the side-stream reactor to assure the heterogeneous diffusion of air, which may require excessive electric energy. However, the mixing in the side-stream reactor is only needed in the micro-aeration period. In a word, there is still a long way to achieve real practical application.

4. Conclusions

SMA is a promising strategy for maintaining systematic stability in AD of high-loaded, easily-hydrolyzed substrates. SMA promoted hydrolysis, acidification, and methanogenesis processes and improved digestion performance. The participation of trace oxygen in SMA

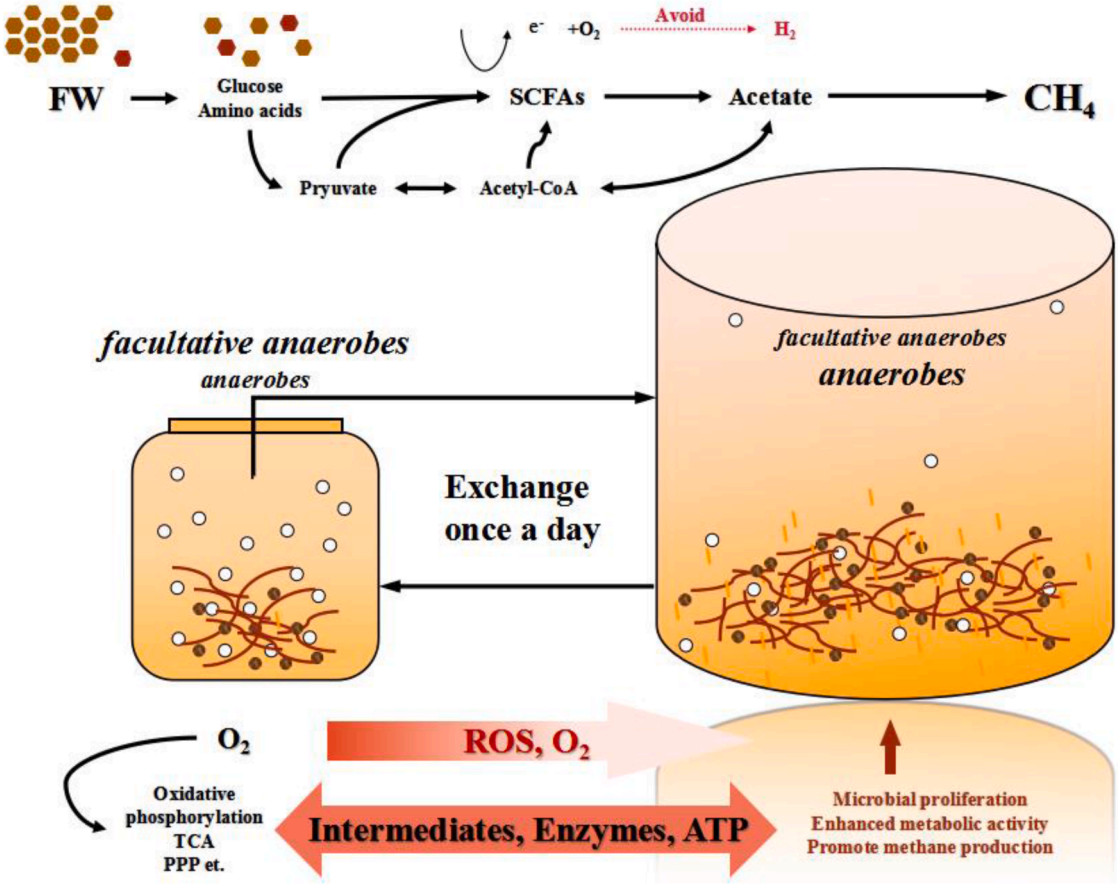


Fig. 7. Concept diagram of the advantages of SMA.

promoted the conversion of SCFA to acetate, which avoids excessive accumulation of FAs. SMA enriched *Methanosaeta* and acetoclastic methanogenesis becomes the main methanogenic pathway. Additional, SMA effectively controlled the negative effects of ROS and helped Methanosaeta occupy most of the ecological niches. Through cycling, SMA exchanged favorable intermediates, enzymes, and ATP, which promoted microbial metabolism and proliferation.

Declaration of Competing Interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper and all authors have seen and approved the final version of the manuscript. They warrant that the article is the authors' original work, has not been published prior and is not under consideration for publication elsewhere.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to acknowledge the financial support for this work provided by the National Natural Science Foundation of China (No. 51208531) and Natural Science Foundation Project of Chongqing (cstc2017jcyjAX0173). Thanks for Mr. Zhang Yong, an engineer in Tangjiatuo WWTP to kindly provide activated sludge and digested sludge. Thanks to Majorbio Cloud Platform for providing technical support on macrogenomics.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2022.118850.

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